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LENS ARRAY BASED IMAGING SYSTEM WITH IMPROVED FIELD OF VIEW

Abstract

An imaging system and method includes a microscope for observing a sample. A light source is arranged to generate light along an optical path of the microscope. A lens array is positioned in the optical path of the microscope, the lens array having a plurality of lenses each with a lens optical axis, the lens optical axes positioned to each capture a separate field of view of the sample. At least one detector is configured to detect the separate fields of view from the lenses. The imaging system is configured to mosaic the detected separate fields of view from each lens to generate an image of the sample.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a U.S. National Phase of International Application No. PCT/US2023/019360 entitled “LENS ARRAY BASED IMAGING SYSTEM WITH IMPROVED FIELD OF VIEW,” and filed on Apr. 21, 2023. International Application No. PCT/US2023/019360 claims priority to U.S. Provisional Patent Application No. 63/334,399 filed on Apr. 25, 2022. The entire contents of each of the above-listed applications are hereby incorporated by reference for all purposes.

FIELD OF THE TECHNOLOGY
[0002] The subject disclosure relates to imaging systems, and more particularly to microscope systems in mesoscale.
BACKGROUND OF THE TECHNOLOGY
[0003] Microscope technology requires the balancing of a number of important design considerations. For example, it should be appreciated that the microscope should be able to capture a large enough field of view (FOV) to properly image an entire sample simultaneously. Obtaining a sufficient FOV also should be balanced against other considerations, such as the image quality, cost, and speed. For example, obtaining a larger FOV can impact the spatial or temporal resolution of the microscope. Therefore, there is an urgent ongoing need in the art to improve existing microscopes, including providing a microscope that allows for large FOV without sacrificing the spatial or temporal resolution.
SUMMARY
[0004] In light of the needs described above, in at least one aspect, the subject technology relates to a microscope that utilizes a lens array to realize a large FOV without tradeoffs on spatial and temporal resolution.
[0005] In at least one aspect, the subject technology relates to an imaging system including a microscope for observing a sample. A light source is arranged to generate light along an optical path of the microscope. A lens array is positioned in the optical path of the microscope, the lens array having a plurality of lenses each with a lens optical axis, the lens optical axes positioned to each capture a separate field of view of the sample. At least one detector is configured to detect the separate fields of view from the lenses. The imaging system is configured to mosaic the detected separate fields of view from each lens to generate an image of the sample.
[0006] In some embodiments, the lens array is a gradient index (GRIN) lens array. A scanner can be positioned in the optical path between the light source and the lens array and configured to move to change the angle of incidence of light on the lens array. In some embodiments, a dichroic mirror is positioned in the optical path between the scanner and the at least one detector to reflect light returning from the sample to the at least one detector. An additional lens array can be positioned between the dichroic mirror and the at least one detector, the additional lens array configured to focus light from each lens in the GRIN lenses separately to the at least one detector.
[0007] In some embodiments, the at least one detector is a plurality of detectors, the plurality of detectors including one detector for each GRIN lens and corresponding to one pixel of the image of the sample. In some cases, a mask pinhole array is positioned near a focal point between the additional lens array and the detector, the mask pinhole array having pinholes corresponding to the focal point of each GRIN lens, the mask pinhole array blocking scattered light around the pinholes.

In some embodiments, the imaging system further comprises a shield which blocks scattered light around each detector, the shield defining pinholes therethrough corresponding to a center of one of the detectors. The system can include a second additional plurality of lenses within the shield, including one lens corresponding to each of the detectors within the shield to improve the detection efficiency.

[0008] In some embodiments, the scanner is positioned and configured such that movement of the scanner further changes the angle of light returning from the sample to the at least one detector. In some cases, the GRIN lens array includes seven separate GRIN lenses, including a central GRIN lens and six outer GRIN lenses placed around the perimeter of the central lens.

[0009] In some embodiments, a pinhole is positioned near the at least one detector, wherein the at least one detector is a single detector. In some cases, a lens is positioned between the detector and the GRIN lens array and configured to focus signals returning through the GRIN lens array through the pinhole and onto the single detector. In that case, the imaging system is configured to use other technologies, for example, time multiplexing such that the single detector collects signals from the GRIN lenses sequentially. In some embodiments, a dichroic mirror is positioned in the optical path between the scanner and the sample, the at least one detector configured to reflect light returning from the sample to the at least one detector. In some cases, the lens optical axes are arranged in parallel.

[0010] In at least one aspect, the subject technology relates to an imaging system including a microscope for observing a sample. A light source is arranged to generate light along an optical path of the microscope. A GRIN lens array is positioned in the optical path of the microscope, the GRIN lens array comprising a plurality of lenses each with a lens optical axis. The lens optical axes are arranged in parallel, and each GRIN lens is positioned to capture a separate field of view of the sample. At least one detector is configured to detect the separate fields of view from the GRIN lenses. A control and acquisition system includes a processor and software for generating an image of the sample from signals from the at least one detector. The control and acquisition system are configured to denoise the signals and construct an image of the sample by tiling the separate field of views from each GRIN lens.

[0011] In at least one aspect, the subject technology relates to a method of imaging a sample with a microscope. Light is generated with a light source along an optical path of the microscope. The light is passed through a lens array positioned in the optical path of the microscope, the lens array having a plurality of lenses each with a lens optical axis, the lens optical axes positioned to each capture a separate field of view of the sample. At least one detector detects the separate fields of view from the lenses. The detected separate fields of view are mosaicked from each lens to generate an image of the sample.

[0012] In some embodiments, the lens array is a GRIN lens array. The method can further include moving a scanner to change the angle of incidence of light on the lens array, the scanner positioned in the optical path between the light source and the lens array. In some embodiments, the method includes reflecting light returning from the sample to the at least one detector with a dichroic mirror positioned in the optical path between the scanner and the at least one detector. In some embodiments, the lens optical axes are arranged in parallel such that light is passed through each lens in parallel.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] So that those having ordinary skill in the art to which the disclosed system pertains will more readily understand how to make and use the same, reference may be had to the following drawings.

[0014] FIG. **1** is a block diagram of an imaging system in accordance with the subject technology.
[0015] FIG. **2** is a schematic diagram of an exemplary microscope in accordance with the subject technology.
[0016] FIG. **3** is a schematic diagram of a GRIN lens array which can be used in a microscope in accordance with the subject technology.
[0017] FIG. **4***a* is a schematic diagram of an exemplary optical path through a single GRIN lens with the light illuminated along the optical axis in accordance with the subject technology.
[0018] FIG. **4***b* is a schematic diagram of the optical path of a GRIN lens array with the light illuminated along the optical axis.
[0019] FIG. **4***c* is a schematic diagram of another exemplary optical path through a GRIN lens with the light illuminated at an angle with respect to the optical axis in accordance with the subject technology.
[0020] FIG. **4***d* is a schematic diagram of the optical path of a GRIN lens array with the light illuminated at an angle with respect to the optical axis.
[0021] FIG. **5** is a simplified diagram showing an exemplary detection optical path through part of the microscope of FIG. **2**.
[0022] FIGS. **6***a*-**6***b* are cross-sectional views of various shield and detector array combinations which can be utilized within a microscope in accordance with the subject technology.
[0023] FIG. **7** is a schematic diagram of another exemplary microscope in accordance with the subject technology.
[0024] FIG. **8** is a schematic diagram of another exemplary microscope in accordance with the subject technology.
[0025] FIGS. **9***a*-**9***f* are cross-sectional views of various GRIN lens arrays which can be implemented as part of a microscope in accordance with the subject technology.

DETAILED DESCRIPTION

[0026] The subject technology overcomes many of the prior art problems associated with microscope systems. In brief summary, the subject technology provides a microscope with a lens array (e.g. of gradient-index (GRIN) lenses, glass lenses, plastic lenses, lenses simulated by wavefront modulators such as spatial light modulator (SLM), or other lenses which could focus light) to image a sample. The advantages, and other features of the systems and methods disclosed herein, will become more readily apparent to those having ordinary skill in the art from the following detailed description of certain preferred embodiments taken in conjunction with the drawings which set forth representative embodiments of the present invention. Like reference numerals are used herein to denote like parts. Further, words denoting orientation such as “upper”, “lower”, “distal”, and “proximate” are merely used to help describe the location of components with respect to one another. For example, an “upper” surface of a part is merely meant to describe a surface that is separate from the “lower” surface of that same part. No words denoting orientation are used to describe an absolute orientation (i.e. where an “upper” part must always be at a higher elevation).
[0027] Referring now to FIG. **1**, a block diagram of a simplified imaging apparatus **100** in accordance with the subject technology is shown. The imaging apparatus **100** includes a microscope **102** for viewing a sample **104**. The microscope **102** includes a laser source **106** to illuminate the sample **104** and one or more scanners **108** to change the angle of the light as it passes to the sample **104**. In the embodiment shown, the scanner **108** directs the light through objectives **110** before passing to the sample **104**. Observed signals from the sample **104** are recorded by detectors **112**.
[0028] The microscope **102** is connected to a control and acquisition system **114**, which can handle all necessary input, output, and processing functions of the imaging apparatus **100**. The control and acquisition system **114** can include a controller, an analog-to-digital converter (ADC), software, and other necessary parts to carry out the functions of the imaging system **100**. In this embodiment,

the analog signals from detectors **112** can be amplified by a multi-channel amplifier and then converted to digital signals by data acquisition (DAQ) systems **116** with onboard processing by a field programmable gate array.

[0029] The signals are then delivered to software **118** for further processing, such as denoise and reconstruction. The software **118** is ultimately responsible for mosaicking the fields of view based on the signals of multiple lenses in a lenses array, as discussed in more detail below, to generate images of the sample **104**.

[0030] The control and acquisition system **114** can also include a control module **120** responsible for coordinating multiple devices and transferring signals between them. It should be understood that the control and acquisition system **114** can include other components to carry out the functions described herein, including a processor and software specifically configured to execute instructions in accordance with the functions of the control and acquisition system **114**. For example, the control and acquisition system **114** can include several acquisition/control cards that have multiple I/O ports which can be used to control and synchronize devices. Off-the-shelf software can be used to control and synchronize the data acquisition process, as is realized in traditional scanning microscope systems such as laser scanning confocal microscopes or two-photon microscopes. Based on this, for multi-channel data acquisition, the number of channels of the software **118** will be updated to the one that is needed.

[0031] Referring now to FIG. **2**, an exemplary microscope in accordance with the subject technology is shown. The microscope **200** can function similar to the microscope **102**, and be used as part of a similar optical imaging apparatus **100**, except as otherwise shown and described. In general, the microscope **200** utilizes a GRIN lens array **202** in an arrangement to replace a typical microscope objective, allowing for a large FOV without sacrificing spatial and temporal resolution. This is realized by combining scanning and descanning techniques with the GRIN lens array **202**, to separate the signal from each lens **204***a*-**204***g* (generally **204**) within the GRIN lens array **202**. The structure of the microscope **200** is simplified for showing purpose and conjugate planes are indicated using like reference numerals (plane **206** and plane **208**).

[0032] A sample **210** is positioned on a substrate for viewing with the microscope **200**. A light source **212**, which could be a visible laser, femtosecond laser, LED, or the like is arranged to generate light along an optical path **214** of the microscope **200**. Note that while one optical path **214** is described for simplicity, it will be detailed below that the GRIN lens array **202** actually utilizes a number of separate optical paths for each lens **204** in the array **202**. The light is spatially filtered by a pinhole **216** to generate a point light source, which will later be imaged onto the sample **210**. Other suitable means could also be used to generate the point light source, such as optical fibers. After collimating and expanding through lenses **218**, **220**, the light will pass through a dichroic mirror **222** (DM) and then be reflected by scanners **224**. Notably, the DM **222** could be replaced by other beam splitters, particularly if aimed to record signals that are non-fluorescent.

[0033] Next, the plane **206** of scanners **224** is relayed to the back-aperture plane **206** of the GRIN lens array **202** by using two lenses **226**, **228**. Light will be focused onto the sample **210** through the GRIN lenses array **202**. Based on the number "n" of GRIN lenses **204** that are used in the array **202**, there will be "n" focuses generated simultaneously on the sample plane **208**, as discussed in more detail below. The returning light passes back through the GRIN lens array **202** and is redirected by the scanner **224**. The DM **222** splits the returning light from the original optical path **214**, the light passing through another lens array **234** and then a pinhole array **232** before receipt by the detector array **230**. The final image generated from the microscope **200** can be formed by combining the images from the light collected from the individual lenses **204** of the GRIN lens array **202**.

[0034] The GRIN lens array **202** used in the microscope **200** is composed of multiple GRIN lenses **204**, as shown in FIG. **3**. In the example shown, the GRIN lens array **202** uses seven separate lenses **204** each with a circular cross section. The lenses **204** are arranged with a central lens **204***d*, and the remaining lenses **204***a*-**204***c*, **204***e*-**204***g* uniformly around the perimeter of the central lens **204***d*. The GRIN lenses **204** are arranged next to each other with their optical axes positioned in parallel. The GRIN lenses **204** are aimed to be arranged in a way that the individual lens is responsible for individual FOV, which could be located at any given position in a three-dimensional space. Thus, for another example the GRIN lenses **204** could also be placed at any given position in a three-dimensional matrix, randomly and separately. The pitch of the GRIN lenses **204** are set as about ¼+½*m where m is an integer. Thus, the GRIN lenses **204** are used to focus or collect light instead of relaying the sample plane **208** beneath the GRIN lens array **202** to its top.

[0035] The microscope **200** is designed to allow individual GRIN lens **204** to focus or collect light from their own field of view (FOV). Exemplary light **406** representing incident beams for lenses **204** are shown in FIGS. **4***a*-**4***d* and discussed in more detail below. This creates at least two advantages. First, by mosaicking the light **406** from the individual FOVs of the different lenses **204**, the total FOV achieved by the system is s*n, where's is the FOV of individual GRIN lens **204** and n is the quantity of GRIN lenses **204** used in the array. Thus, the total FOV can be enlarged either by increasing the diameter(s) or quantity (n) of GRIN lenses **204**. Second, because the light collection ability depends on a single lens, the spatial resolution equals to that of individual GRIN lenses **204**, which is determined by its numerical aperture (NA). Therefore, unlike conventional microscope objectives, the spatial resolution is not reduced when enlarging the FOV. While the GRIN lens array **202** has been found to be particularly advantageous, it should be understood that another lens array which functions similar to the GRIN lens array **202** described herein could also be used.

[0036] The scanners **224** of the microscope **200** are designed to move the focuses by changing the angles of light when it incidents on GRIN lenses **204**. This process is known as scanning. The scanners **224** can be galvanometer scanners, resonant scanners, spatial light modulator (SLM), digital micromirror device (DMD), micro-electro-mechanical system (MEMS), acousto-optic deflectors (AOD) or other devices which are capable of changing the direction of light. Light beams directed to each GRIN lens can be controlled by the same or separate scanners **224**. The effect of scanning on light within the system is illustrated in FIGS. **4***a*-**4***d*. More particularly, FIG. **4***b* shows light **406** passing through the GRIN lenses **204** in a first orientation, parallel to the central axis **408** of the GRIN lens array **202**. FIG. **4***a* shows an exemplary path of the light **406** of FIG. **4***b* passing through an individual lens **204** within the GRIN lens array **202**. FIG. **4***d* shows light **406** passing through the GRIN lens array **202** at an oblique angle (with respect to the central axis **408**) due to scanning. FIG. **4***c* shows an individual lens **204** of the array **202** shown in FIG. **4***d*.

[0037] The scanning process can employ other known scanning strategies used in confocal or two-photon (2P) systems, specific or all locations across the FOV of individual GRIN lenses **204** will be reached. Since the scanning through multiple GRIN lenses **204** is accomplished simultaneously, the time spent on scanning of the whole FOV equals to that on individual ones. That means, if the time spent on scanning the FOV(s) of single GRIN lens **204** is t, the time spent on scanning the whole FOV (n*s) is also t, as the light beams **406** are scanning simultaneously. Thus, the temporal resolution is not reduced when enlarging the FOV by using multiple GRIN lens **204**.

[0038] Notably, various scanning processes can be employed by the microscope **200**, including random, raster, or other customized processes. In the embodiment shown, during scanning, all parts of the microscope **200** except for scanners **224** remain immobilized, and there is no movement between the sample and objective. Alternatively, the microscope **200** can use sample scanning instead of light beam scanning. In that case, the direction of light beam that incidents onto GRIN lenses **204** is fixed, while the sample **210** or lens array **202** is moving, for example, in a 2D, or 3D manner.

[0039] Referring now to FIG. **5** a simplified schematic diagram shows an exemplary part of the path of signals from the sample **210** through the microscope **200**. The responding signals from individual focus are collected by corresponding GRIN lenses **204**. These signals could be fluorescence or other light induced signals, as well as reflected or transmitted light. Here, fluorescent signals are used as an example. In FIG. **5**, signals **500** (solid lines) represent lights from the center of individual FOVs, while signals **502** (dotted lines) represents lights from another location in the same FOVs.

[0040] Collected fluorescent signals are then relayed back to the scanner **224**, i.e. the back-aperture plane **206** of GRIN lenses array **202** will be relayed back to the conjugate scanner plane **206**. After propagating through the scanners **224**, the locations, propagating directions, and sizes of the beams from each GRIN lens **204** are fixed yet separatedError! Reference source not found. Particularly, for individual lenses **204**, lights from different locations coincide after scanner **224**, (i.e. light **500**, **502** coincide after the scanner **224**). This means the propagating direction will not change during scanning. This process is called descanning. More importantly, signals from individual GRIN lenses **204** are separated after descanning. Therefore, a second lens array **234** includes a number of individual lenses each focusing a respective beam separately. The diameter of lenses in the second lens array **234** matches the size of the image of individual GRIN lens **204**.

[0041] As the location and direction of lights from each GRIN lens **204** are fixed after descanning, the location of focus is then fixed on the image plane **208** of the second lens array **234**. This means, for each GRIN lens **204**, signals on the image plane **208** always reaches the same point during scanning. For example, the image location of the light from each GRIN lens is fixed at a corresponding point **238***a*-**238***g* (generally **238**) while scanning. A mask pinhole array **232** is placed at the focus point in the example shown. The mask pinhole array **232** contains a pinhole array with different pinholes located at different focus, corresponding to the lenses **204**. The pinhole array **232** will reject light scattered from sample locations other than its conjugated focus, similar to pinholes used in confocal microscope systems, which will improve the sectioning ability and signal-to-noise ratio (SNR). Furthermore, using pinholes reduces potential crosstalk between individual GRIN lenses **204**. The detector array **230** (e.g. photo-sensitive array) is located behind the mask pinhole array **232** to record optic signals. Signals from each GRIN lens **204** are recorded separately in the detector array **230**. Of note, signals could also be collected using fibers and then be delivered to different pixels, or different detectors. The final image is a combination of these collected signals.

[0042] If n GRIN lenses **204** are used in the microscope **200** (relay lenses aside) there will be n parallel optical axes (one for each GRIN lens **204**). For each pairing of a GRIN lens **204** and pinhole, centers are formed located on their own optical axis. In FIG. **5**, only two optical axes **504** of two GRIN lenses **204** are shown for simplicity. If the relay lenses are well constructed, the off-axis aberrations are significantly less when compared to conventional methods under the same FOV. It should be understood that the pinhole **216** and pinhole arrays **232** shown herein are recommended and advantageous, but are not absolutely necessary if there are no needs to improve the section abilities or SNR. This imaging method can also be combined with other scanning strategies such as line scanning, multi-foci scanning and others to further improve the speed, image quality or other parameters.

[0043] Furthermore, when individual detectors within the detector array **230** are close to each other, such as these pixels of EMCCD, or multianode PMT, signal crosstalk between pixels or detectors could exist. This crosstalk could be decreased by increasing the distance between focuses on the image plane. Another way is to use a shield with small chambers to keep the signal away from reaching other detectors (or pixels), as shown in FIGS. **6***a-b*.

[0044] Referring now to FIGS. **6***a-b*, embodiments of a detector array with individual detectors **600**, or a detector with individual pixels **600**, each utilizing a shield **602** are shown. On top of this shield **602**, there are multiple pinholes **604**, including one for each detector **600** at a center of the corresponding detector **600**. Light **606** passes through individual pinholes **604** corresponding to the center of each detector **600**, with the shield **602** forming small chambers around each detector **600**. Each detector **600** can correspond to one pixel in the final image of the sample **210**. Taking the multinode PMT as an example, the pixel size can be 6 mm. The distance between individual chambers **608** formed by the shield **602** is thus set as 6 mm while the width of chamber **608** is, for example, 2 mm. Because light signals are restricted within individual chambers **608**, theoretically no signals will fall onto the junction of adjacent pixels **600** and the crosstalk is then reduced. In addition, lenses **610** can also be placed in front of detectors to improve the collection efficiency of light, as shown in FIG. **6***b*. An alternative method of reducing crosstalk between detectors **600** is, as previously mentioned, using fibers to deliver the signal from each GRIN lens **204** to the separated detectors **600**. Furthermore, this system can also be employed in conjunction with movement between the sample and objective, which will further enhance the FOV without altering the GRIN lens array **202**.

[0045] Referring now to FIG. **7**, another exemplary embodiment of a microscope **700**, in accordance with the subject technology, is shown. The microscope **700** includes similar components, and can function similarly, to the microscope **200**, except as otherwise shown and described. In the microscope **200**, a second lens array **234** and detector array **230** of multiple detectors are adopted to realize large FOV imaging without tradeoffs on spatial and temporal resolution. By contrast, in the microscope **700**, the second lens array **234** is replaced by a single lens **702** and the detector array **230** is replaced by a single detector **704**. The single lens **702** and single detector **704** are used to focus and record signals after descanning. In this microscope **700**, light from all GRIN lenses **204** will be focused to the same location **706** (i.e. representing overlapping light from the difference GRIN lenses **204**). The signals from each GRIN lens **204** are focused through a single pinhole **708** and combined on the detector **704**. In this case, other strategies, for example, time multiplexing methods can be used, to separate these signals. To realize time multiplexing, other devices can be incorporated such as a spatial light modulator, or separate scanners, to control the on and off status of individual GRIN lenses **204**. In other words, signals from individual GRIN lenses **204** will be collected sequentially by the detector **704**. This strategy allows large FOV without compromising spatial resolution. The temporal resolution, however, might be reduced.

[0046] Referring now to FIG. **8**, another exemplary embodiment of a microscope **800**, in accordance with the subject technology, is shown. The microscope **800** includes similar components, and can function similarly, to the microscope **200**, except as otherwise shown and described. Unlike the microscope **200**, there is no descanning in the microscope **800**. This allows less signal loss as detector array **230** is placed closer to the sample **210**, making it more suitable for detecting weak signals. In this embodiment, lights from sample **210** are collected and then directed to a second lens **802** and lens array **804**, with a dichromatic mirror **222** or a beam splitter. The back aperture of the GRIN lens array **202** is then relayed to a plane **206** where the second lens array **804** is located (line planes being marked with like reference numbers). The lens array **804** will focus lights from GRIN lenses **204** into detectors within the detector array **230**. Different from the microscope **200** where descanning is adopted, in microscope **800**, for each GRIN lens **204**, imaging points from different sample locations will locate at different places on the image plane **208**. This means, when scanning, the focusing point will move (or scan) on the detector array **230**. Exemplary indicators **808***a*-**808***g* are shown on the image plane **208** corresponding to the respective GRIN lenses **204***a*-**204***g*. Dotted lines **810** through the lens array represent the change of signal angles during scanning (compared to solid lines **812**), with dots **814** on the image plane representing exemplary signal movement during scanning. Thus, pinholes cannot be used to reject the out-of-focus signals and the size of each detector in the detector array **230** should be large.

[0047] Referring now to FIGS. **9***a*-**9***f*, exemplary cross sections of GRIN lens arrays **900***a*-**900***f* (generally **900**) are shown, the arrays **900** each comprised of a plurality of individual GRIN lenses **902***a*-**902**-*f* (generally **902**). It should be understood that the arrays **900** are exemplary only, and other arrangements could also be used within a microscope designed in accordance with the subject technology. In some examples, the GRIN lenses arrays **900** can be arranged in various shapes, such

as circular (**900***a*, **900***d*), square (**900***b*, **900***c*, **900***e*, **900***f*), or other shapes, with the total number of GRIN lenses **902** varying as needed. The cross section of each individual GRIN lens **902** could also be different shapes, including circular (**902***a*, **902***b*, **902***e*, **902***f*), rectangular (**902***c*), square, hexagonal (**902***d*), or others. As shown in FIGS. **9***e*-**9***f*, the array **900** can also include GRIN lenses **902***e*, **902***f* with varying size, such as a first grouping of larger lenses of a first size and a second grouping of lenses of a smaller size. The relative positions between individual GRIN lenses are changeable. They can be placed separately instead of being adjacent to each other. For example, multiple lenses can be placed to focus on different regions within the sample to record the signals from these regions simultaneously.
[0048] The objective (e.g. GRIN lens arrays) of the system is changeable. In some cases, several objectives of different parameters, such as size, numerical aperture, and number of lenses can be used. Further, the system can process the light using a number of separate channels, including up to 64 channels, with one channel for each lens in the lens array. A rotary stage can also be used to shift between different arrays as needed.
[0049] Notably, while GRIN lenses are discussed throughout this disclosure, and have been found to be one advantageous type of lens array which can be used as part of the systems described herein, it should be understood that other lens arrays may also be sued. For example, the systems described herein could use glass lenses, plastic lenses, lenses simulated by wavefront modulators such as spatial light modulators, or other lenses which could focus light, instead of a GRIN lens array.
[0050] All orientations and arrangements of the components shown herein are used by way of example only. Further, it will be appreciated by those of ordinary skill in the pertinent art that the functions of several elements may, in alternative embodiments, be carried out by fewer elements or a single element. Similarly, in some embodiments, any functional element may perform fewer, or different, operations than those described with respect to the illustrated embodiment. Also, functional elements shown as distinct for purposes of illustration may be incorporated within other functional elements in a particular implementation.
[0051] While the subject technology has been described with respect to preferred embodiments, those skilled in the art will readily appreciate that various changes and/or modifications can be made to the subject technology without departing from the spirit or scope of the subject technology. For example, each claim may depend from any or all claims in a multiple dependent manner even though such has not been originally claimed.

Claims

1. An imaging system including a microscope for observing a sample comprising: a light source arranged to generate light along an optical path of the microscope; a lens array positioned in the optical path of the microscope, the lens array comprising a plurality of lenses each with a lens optical axis, the lens optical axes positioned to each capture a separate field of view of the sample; and at least one detector configured to detect the separate fields of view from the lenses, wherein the imaging system is configured to mosaic the detected separate fields of view from each lens to generate an image of the sample.
2. The imaging system of claim 1, wherein the lens array is a gradient index (GRIN) lens array.
3. The imaging system of claim 2, further comprising a scanner positioned in the optical path between the light source and the lens array and configured to change the angle of incidence of light on the lens array.
4. The imaging system of claim 3, further comprising a dichroic mirror positioned in the optical path between the scanner and the at least one detector to reflect light returning from the sample to the at least one detector; and further comprising an additional lens array positioned between the dichroic mirror and the at least one detector, the additional lens array configured to focus light from

each lens in the GRIN lenses separately to the at least one detector.

5. (canceled)

6. The imaging system of claim 4, wherein the at least one detector is a plurality of detectors, the plurality of detectors including one detector for each GRIN lens.

7. The imaging system of claim 6, wherein each GRIN lens corresponds to one subfield-of-view of the image of the sample.

8. The imaging system of claim 6, further comprising a mask pinhole array positioned near a focus between the additional lens array and the detector, the mask pinhole array having pinholes corresponding to the focus of each GRIN lens, the mask pinhole array blocking scattered light around the pinholes.

9. The imaging system of claim 3, further comprising a dichroic mirror positioned in the optical path between the scanner and the at least one detector to reflect light returning from the sample to the at least one detector, wherein the imaging system further comprises a shield which blocks scattered light around each detector, the shield defining pinholes therethrough corresponding to a center of one of the detectors, and a second additional plurality of lenses within the shield, including one lens corresponding to each of the detectors within the shield.

10. (canceled)

11. The imaging system of claim 3, wherein the scanner is positioned and configured such that movement of the scanner further changes the angle of light returning from the sample to the at least one detector.

12. The imaging system of claim 2, wherein the GRIN lens array includes seven separate GRIN lenses, including a central GRIN lens and six outer GRIN lenses placed around the perimeter of the central lens.

13. The imaging system of claim 3, further comprising: a pinhole positioned near the at least one detector, wherein the at least one detector is a single detector; and a lens positioned between the detector and the GRIN lens array and configured to focus signals returning through the GRIN lens array through the pinhole and onto the single detector, wherein the imaging system is configured to use a time multiplexing technique such that the single detector is capable of distinguishing the signals from the GRIN lenses.

14. The imaging system of claim 3, further comprising a beam splitter positioned in the optical path between the scanner and the sample.

15. The imaging system of claim 13, wherein the beam splitter comprises a dichroic mirror configured to reflect light returning from the sample to the at least one detector.

16. (canceled)

17. An imaging system including a microscope for observing a sample comprising: a light source arranged to generate light along an optical path of the microscope; a gradient index (GRIN) lens array positioned in the optical path of the microscope, the GRIN lens array comprising a plurality of lenses each with a lens optical axis, the lens optical axes arranged in parallel and each GRIN lens positioned to capture a separate field of view of the sample; at least one detector configured to detect the separate fields of view from the GRIN lenses; and a control and acquisition system, including a processor and software for generating an image of the sample from signals from the at least one detector, the control and acquisition system configured to denoise the signals and construct an image of the sample by mosaicking the separate field of views from each GRIN lens.

18. A method of imaging a sample with a microscope comprising: generating light, with a light source, along an optical path of the microscope; passing the light through a lens array positioned in the optical path of the microscope, the lens array comprising a plurality of lenses each with a lens optical axis, the lens optical axes positioned to each capture a separate field of view of the sample; detecting, with at least one detector, the separate fields of view from the lenses; and mosaicking the detected separate fields of view from each lens to generate an image of the sample.

19. The method of claim 18, wherein the lens array is a gradient index (GRIN) lens array.

20. The method of claim 19, further comprising moving a scanner to change the angle of incidence of light on the lens array, the scanner positioned in the optical path between the light source and the lens array.
21. The method of claim 20, further comprising directing light returning from the sample to the at least one detector with beam splitter positioned in the optical path between the scanner and the at least one detector.
22. The method of claim 21, wherein the beam splitter comprises a dichroic mirror.
23. The method of claim 18, wherein the lens optical axes are arranged in parallel such that light is passed through each lens in parallel.